Practical Project 2

Image-to-Prompt

Recovering prompts that re-render LCM target images

Generative AI | 2025/2026 — 2nd Semester | University of Coimbra (FCTUC)

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Project at a Glance

1

Goal — invert images back to prompts

Given a fixed text-to-image generator and a target image, recover prompts that re-render images close to the target.

2

Top-3 prompts per target

For each of the six target images, return a ranked list of three candidate prompts and their LCM-rendered reconstructions.

3

Image-side evaluation

Judge prompts by what they generate, not by fluency: CLIP image-image similarity, LPIPS, and pixel MSE.

4

Open methodology

Captioning, retrieval, prompt sampling, LLM refinement, gradient-based optimisation, soft-prompt inversion — free choice or combinations.

Problem Statement

Image-to-prompt inversion under a fixed latent diffusion generator

Given

Six target images
produced by SimianLuo/LCM_Dreamshaper_v7, a fast latent consistency model derived from DreamShaper / Stable Diffusion.

Fixed generation setup
Model checkpoint, scheduler family, inference steps, guidance scale, resolution, and per-image random seed are all known.

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Get

The text prompt with a single Methodology.

Recover prompts that, when rendered again with the same LCM and configuration, produce images visually close to the targets.

Not classification, not captioning.
A prompt is judged by the image it produces when re-rendered — not by how well it describes the target in words.

The Inversion Loop

Prompt is judged by the image it re-generates, not by its words

Target image
from tp2-chosen

→

Candidate prompt
your method

→

LCM render
fixed config

→

Image-side metric
CLIP · LPIPS · MSE

Determinism is the key. Same prompt + same configuration → the same image up to numerical noise.
So the search space is text, but the evaluation is in pixels.

Objective

For every target, return three ranked candidate prompts

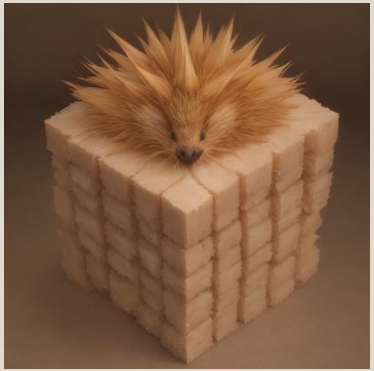
6	3	18	3
target images	prompts per target	rendered candidates	image-side metrics

For each target image, deliver:

Top-3 prompts found by your system, ranked by an image-side similarity criterion.
LCM-rendered images produced from each prompt under the fixed configuration.
Per-candidate metric table with CLIP image-image similarity, LPIPS, and pixel MSE.
Mean and standard deviation of the main metrics aggregated across the full test set.

Target Set

Six images — prompts not provided



Random seed is encoded in the filename — e.g. 7836.png uses seed 7836.

Generator Setup

All candidate prompts must be evaluated under the same configuration

768 × 768	8.0	8	50
RGB resolution	guidance scale	inference steps	LCM origin steps

Model & scheduler

`SimianLuo/LCM_Dreamshaper_v7`
Latent Consistency Model derived from DreamShaper / Stable Diffusion-style models.

Treat the generator as deterministic: same prompt + same configuration → same image (up to numerical noise).

Per-image seed

Encoded in the filename.

```
7836.png → seed 7836
1159_25.png → seed 1159
```

Keep the LCM rendering seed fixed; if your prompt search is stochastic, repeat with multiple optimiser seeds.

Approach Spectrum

Choose one or combine — final output must be discrete text prompts and about ***one final methodology***

Vision-language captioning

BLIP / CLIP-Interrogator-style models propose initial prompt candidates from the image.

Prompt retrieval

Search a prompt corpus by CLIP image embedding, then re-rank by LCM-rendered similarity.

Prompt sampling

Generate templated or LLM-sampled prompts, render, score, keep the best.

LLM-based refinement

Iteratively edit the prompt with an LLM using image-side feedback as the signal.

Discrete prompt optimisation

Gradient-based methods (e.g. PEZ, hard-prompt search) over the tokenizer vocabulary.

Soft-prompt inversion

Optimise embedding tokens, then project back to discrete text for final submission.

Evaluation: Image-Side Metrics

Prompt fluency alone is not sufficient — metrics live in pixels

CLIP image–image

Cosine similarity between target and rendered image embeddings from a fixed CLIP encoder (e.g. openai/clip-vit-large-patch14).

higher is better

LPIPS

Learned perceptual distance between deep features of target and rendered images — sensitive to perceptual structure.

lower is better

Pixel MSE

Mean squared error in pixel space between target and rendered image — a strict, low-level fidelity check.

lower is better

All metrics are computed at the image level, comparing each target with each rendered candidate.

Reporting Protocol

What every target entry and the aggregate must include

Per target image

- the target image itself
- the three recovered prompts, ordered best to worst
- the three rendered images
- a per-candidate metric table

Across the full set

- mean and standard deviation of the main metrics
- if the prompt search is **stochastic**, repeat with multiple seeds **(at least 5)**
- keep the LCM rendering seed fixed across all repetitions
- qualitative analysis of successes, failures, and ambiguous cases

Deliverables

1. **Written report (LNCS format)**
   12 pages excluding references. Sections: introduction, problem description, methodology and approach, experimental setup, analysis of results, conclusion. Portuguese or English.

2. **All code to reproduce results**
   Including model versions, hyperparameters, and the random seeds used by every stochastic component of the pipeline.

3. **Acknowledgement of all sources**
   External code, pretrained components, datasets, and any generative AI tools used during development must be explicitly identified.

Logistics & Constraints

8 Jun	2	12	Indiv.
deadline (2026)	students per group	page limit (LNCS)	final mark

What is judged

- Technical work
- Written description (LNCS report)
- Public defence

Pay attention to

- Clear description of approach and external components
- Reproducible protocol: model versions, hyperparameters, seeds
- Quantitative + qualitative analysis

Search the text. Score the pixels.

Prompt inversion is an optimisation problem under a fixed text-to-image generator.

Connect method

to the concrete LCM setup and the metric you are optimising.

Connect renders

to the qualitative behaviour of your top-3 prompts.

Connect numbers

to the success and failure cases you actually observe.

Good luck.